

How accurate are Bayes factor-based null hypothesis tests? A simulation study

Daniel J. Schad¹, Martin Modrák², & Shravan Vasishth³

¹ Institute for Mind, Brain and Behavior, HMU Health and Medical University, Potsdam,
Germany

² Second Faculty of Medicine, Charles University, Prague, Czech Republic

³ University of Potsdam, Potsdam, Germany

Author Note

Correspondence concerning this article should be addressed to Daniel J. Schad, HMU Health and Medical University, Potsdam, Germany. E-mail: danieljschad@gmail.com. A preprint of this manuscript has been published on arXiv (<https://arxiv.org/abs/2406.08022>). Reproducible code is available from <https://osf.io/3g86r/>. This study was not preregistered.

Abstract

Bayes factor null hypothesis tests provide a viable alternative to frequentist measures of evidence quantification. Bayes factors for realistic data sets in areas like psychology cannot be calculated exactly and require numerical approximations to complex integrals. Crucially, the accuracy of these approximations, i.e., whether an approximate Bayes factor corresponds to the exact Bayes factor, is unknown, and may depend on data, prior, and likelihood. We have recently developed a novel statistical procedure, namely marginal simulation-based calibration (SBC) for Bayes factors, to test whether the computed Bayes factors for a given analysis are accurate. Here, we use marginal SBC for Bayes factors and calibration plots to test for some common cognitive designs, whether Bayes factors are calculated accurately. We use the `bridgesampling`/`brms` packages in R. We run analyses for three commonly used designs in psychology and psycholinguistics: (a) a design with random effects for subjects only, (b) a Latin square design with crossed random effects for subjects and items, but a single fixed-factor, and (c) a Latin square 2x2 design with crossed random effects for subjects and items. We find that Bayes factor estimates turn out accurate in cases when the `bridgesampling` algorithm does not issue a warning message, but can be biased and liberal when a warning message is shown. These results support the use of `brms`/`bridgesampling` for null hypothesis Bayes factor tests in commonly used factorial designs. They also suggest that when a warning message is issued, Bayes factor results should not be trusted. The results show that it is practical to check whether Bayes factors are computed correctly.

Keywords: Bayes factors, Bayesian model comparison, simulation-based calibration, null hypothesis testing, crossed random effects for subjects and items

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Introduction

Bayes factors have emerged as a popular tool for quantifying evidence for hypotheses and comparing models in statistical data analysis (Chow & Hoijsink, 2017; Hoijsink & Chow, 2017; Lee, 2011; Mulder & Wagenmakers, 2016; Vandekerckhove, Rouder, & Kruschke, 2018). Bayes factors offer several advantages, such as their ability to assess the probability of the null hypothesis, incorporate prior information, and provide continuous measures of evidence strength. As a result, Bayes factors are increasingly used in many scientific domains, including psychology, neuroscience, and social sciences (Heck et al., 2023).

However, computing exact Bayes factors is often not feasible, as it involves evaluating complex integrals that cannot be solved analytically, and approximations are necessary. For some restricted families of models and prior distributions, Bayes factors can be converted to relatively well-behaved low-dimensional integrals that are then directly evaluated numerically — an example of this approach is the BayesFactor (Morey & Rouder, 2022) R package. For more general models, we typically need to first estimate posterior distributions for both models with a suitable Markov-chain Monte Carlo (MCMC) sampler and then use the posterior samples to build an approximation to the marginal likelihood in a separate computational step — an example of this approach is the bridgesampling (Gronau, Singmann, & Wagenmakers, 2020) R package. In the former case, the precision of the approximation is limited by the precision of numerical integration, in the latter case the approximation is affected by imprecisions in both the MCMC step and in the marginal likelihood computation. Nevertheless, the quality of these approximations are as yet unclear and remain a concern, as biased approximations could lead to erroneous conclusions about the evidence for hypotheses and the relative strengths of models (Schad, Nicenboim, Bürkner, Betancourt, & Vasishth, 2023).

To investigate the accuracy of Bayes factor estimates, Schad et al. (2023) recently